

UNIT 20

NUCLEAR STRUCTURE



NATURAL RADIOACTIVITY

Natural Radioactivity refers to the spontaneous emission of radiation from certain unstable atomic nuclei, resulting in their transformation into more stable forms. This phenomenon occurs without any external influence and is a property of certain elements. There are three main types of radiation associated with natural radioactivity:

1. Alpha (α) Radiation: Alpha particles are composed of two protons and two neutrons, making them positively charged. They have low penetrating power but can cause significant damage if emitted within the body.

2. Beta (β) Radiation: Beta particles are high-energy electrons (β^-) or positrons (β^+) emitted from the nucleus. They have higher penetrating power compared to alpha particles.

3. Gamma (γ) Radiation: Gamma rays are electromagnetic waves of high energy and very short wavelength. They have the highest penetrating power and are often emitted along with alpha or beta particles.

These types of radiation can interact with matter, ionizing atoms and potentially causing biological damage.

NATURE OF RADIOACTIVE EMISSION

Radioactive emissions involve the release of particles and energy from unstable atomic nuclei in the process of achieving greater stability. This emission occurs spontaneously without external influence. The nature of radioactive emission includes three main types:

1. Alpha Emission: Unstable nuclei emit alpha particles, which consist of two protons and two neutrons. These particles are relatively large and positively charged, resulting in low penetration power. Alpha emission reduces the atomic number of the nucleus, effectively transforming it into a different element.

2. Beta Emission: During beta emission, a neutron in the nucleus is transformed into a proton, emitting a high-energy electron (beta-minus, β^-) or a positron (beta-plus, β^+). Beta particles are smaller and more penetrating than alpha particles, and they can change the atomic number of the nucleus.

3. Gamma Emission: After alpha or beta emission, the nucleus might still be in an excited state. To reach a stable energy level, it releases excess energy in the form of gamma radiation. Gamma rays are high-energy electromagnetic waves with very short wavelengths, possessing the highest penetrating ability.

RELATIVE IONIZING EFFECTS OF RADIOACTIVE EMISSION

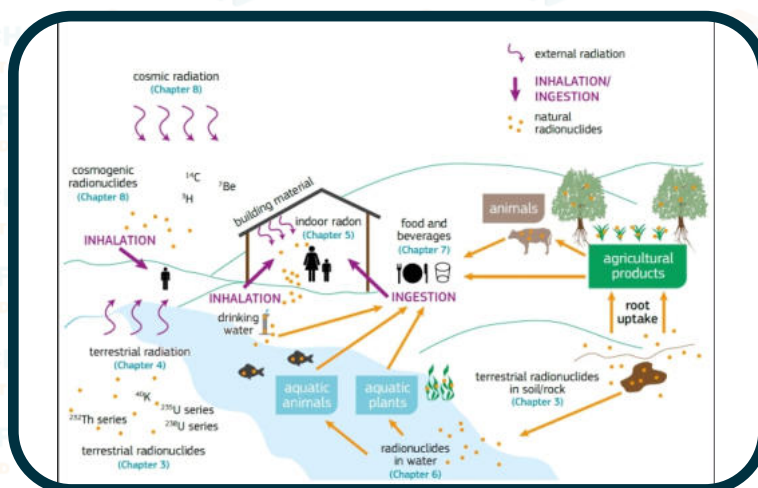
The relative ionizing effects of radioactive emission refer to the ability of different types of radiation to cause ionization in the matter they interact with.

IONIZATION

Ionization is the process of removing electrons from atoms, creating charged particles (ions). Different types of radioactive emissions have varying degrees of ionizing effects.

PENETRATING POWER

Penetrating power refers to the ability of radiation, such as alpha, beta, or gamma rays, to travel through matter and penetrate barriers. It is determined by the type and energy of radiation, with alpha particles having low penetrating power, beta particles having moderate, and gamma rays having high penetrating power.



NUCLEAR

TRANSMUTATIONS

Parent Nucleus: The parent nucleus refers to an unstable atomic nucleus that undergoes radioactive decay or disintegration, emitting particles and energy to transform into a more stable configuration.

Daughter Nucleus: The daughter nucleus is the result of the radioactive decay of a parent nucleus. It's formed after emission of particles or energy during the decay process, leading to a change in the atomic and/or mass number.

Radioactive Disintegration: Radioactive disintegration is the spontaneous process in which an unstable atomic nucleus transforms into a more stable state by emitting radiation such as alpha or beta particles, or gamma rays. This emission is a characteristic feature of radioactive materials aiming to achieve greater stability over time.

ALPHA DECAY

Alpha decay is a type of radioactive decay in which an unstable atomic nucleus emits an alpha particle, which consists of two protons and two neutrons. This reduces the atomic number by 2 and the mass number by 4. An example is the decay of uranium-238 to thorium-234:

Uranium-238 (parent nucleus) $\rightarrow$ Thorium-234 (daughter nucleus) + Alpha particle (helium nucleus)

BETA DECAY

Beta decay involves the emission of a beta particle from an unstable atomic nucleus. Beta particles can be either electrons (beta-minus, β^-) or positrons (beta-plus, β^+). In beta-minus decay, a neutron is transformed into a proton and an electron is emitted. In beta-plus decay, a proton becomes a neutron, and a positron is emitted. An example of beta-minus decay is the decay of carbon-14:

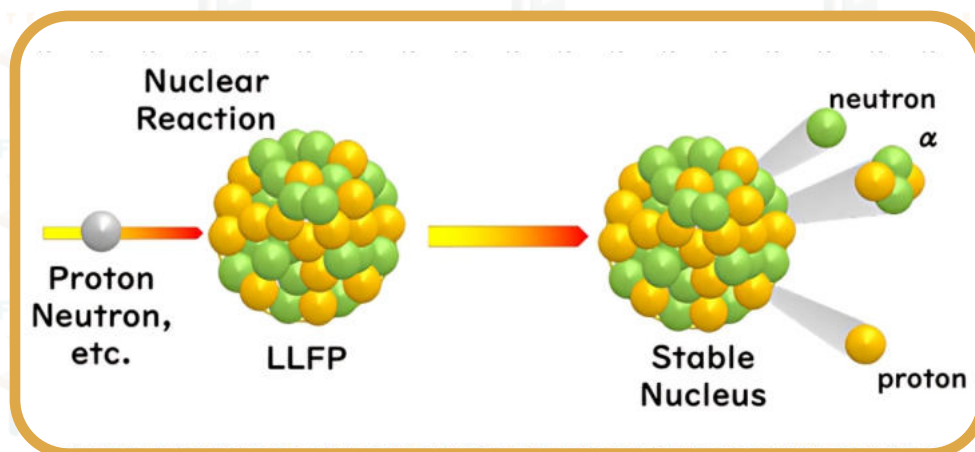
Carbon-14 (parent nucleus) $\rightarrow$ Nitrogen-14 (daughter nucleus) + Beta-minus particle (electron)

BETA+ DECAY

Beta-plus decay involves the emission of a positron, which is a positively charged electron, along with a neutrino. It occurs when a proton is transformed into a neutron. An example is the decay of fluorine-18:

Fluorine-18 (parent nucleus) $\rightarrow$ Oxygen-18 (daughter nucleus) + Beta-plus particle (positron) + Neutrino

These decay processes are ways for unstable nuclei to achieve greater stability by adjusting their proton-neutron ratios and releasing excess energy in the form of radiation.

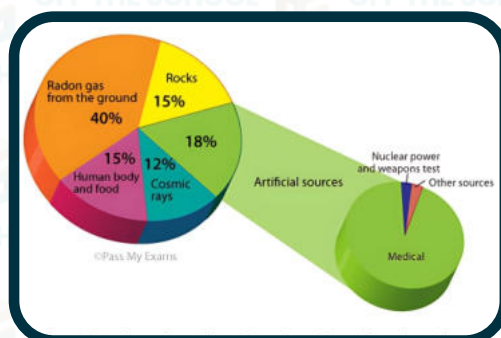


Background radiation refers to the continuous low-level ionizing radiation present in the environment from natural and human-made sources. It includes cosmic rays, radioactive elements in soil, rocks, and buildings, as well as medical procedures, nuclear testing, and industrial activities.

GEIGER – MULLER (GM) TUBE

The Geiger-Muller (GM) tube, also known as a Geiger counter, is a radiation detection device that can demonstrate the nature of observing the count rate of radioactive disintegration. It works based on the principles of ionization caused by radioactive particles. Here's how it demonstrates this nature:

- 1. Ionization and Detection:** Conversion of a radioactive particle's energy into an electrical pulse when it interacts with gas molecules in a Geiger-Muller tube.
- 2. Amplification:** Increasing the strength of the electrical pulse generated by ionization for easier detection.
- 3. Counting Rate:** The number of electrical pulses detected per unit of time, reflecting the rate of radioactive particle interactions in a Geiger counter.
- 4. Nature of Radioactive Decay:** The random and spontaneous process by which unstable atomic nuclei emit particles or energy to attain stability.
- 5. Visual and Audible Indication:** Display or sound cues in a Geiger counter that signal the detection of radioactive particles, providing observable evidence of radioactivity.



HALF-LIFE

The refers to the time it takes for half of a sample of that material's unstable atomic nuclei to undergo radioactive decay and transform into a more stable form. In other words, it's the duration it takes for the initial quantity of radioactive substance to reduce by half due to the emission of particles or radiation. The concept of half-life is crucial in understanding the rate of decay and the behavior of radioactive substances over time. Different radioactive materials have unique half-lives, ranging from fractions of a second to millions of years, and this property is utilized in various fields such as radiometric dating, medical imaging, and nuclear energy.

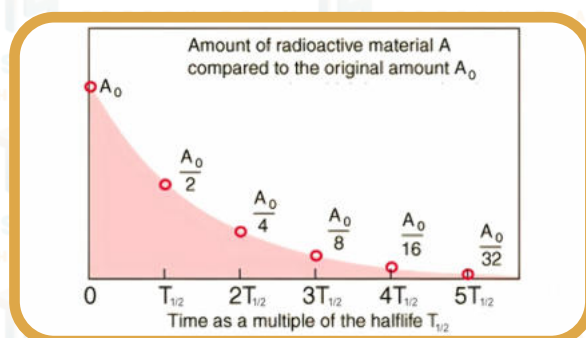
ISOTOPE	HALF-LIFE
Carbon-14	5,730 years
Uranium-238	4.468 billion years
Potassium-40	1.25 billion years
Radon-222	3.8 days
Cobalt-60	5.27 years
Iodine-131	8.02 days
Strontium-90	28.8 years
Technetium-99m	6 hours

RADIOACTIVITY

Radioactivity is the spontaneous emission of particles or radiation from unstable atomic nuclei, aiming to achieve stability, and is a fundamental property of certain elements.

Radioactive dating:

Radioactive dating is a method of determining the age of objects or materials by measuring the decay of radioactive isotopes present within them, using the known half-lives of these isotopes as a clock to estimate the time since their formation.

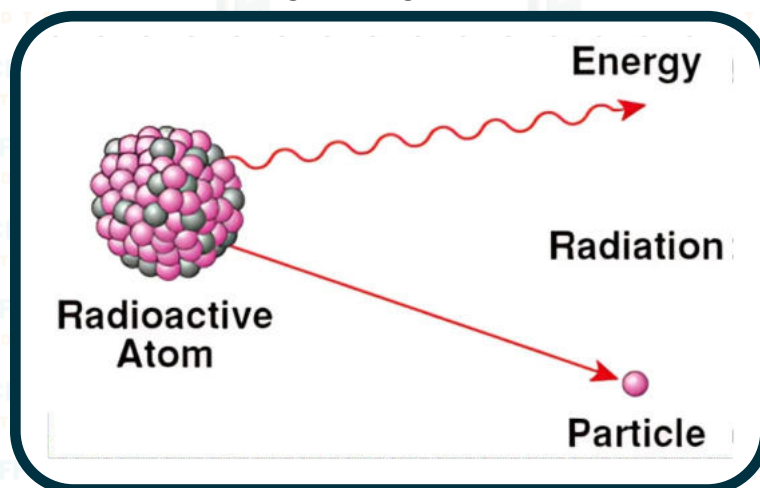


RADIOACTIVE ISOTOPES

Radioactive isotopes, or radioisotopes, are variants of chemical elements with the same number of protons but different numbers of neutrons in their nuclei, making them unstable and prone to emit radiation during radioactive decay.

Example:

An example is carbon-14 used in radiocarbon dating, where the decay of carbon-14 to nitrogen-14 is used to determine the age of organic materials.



APPLICATIONS OF RADIOISOTOPES

MEDICINE

- 1. Medical Imaging:** Radioisotopes like technetium-99m are used in diagnostic imaging techniques like gamma scintigraphy to visualize internal organs and detect abnormalities.
- 2. Cancer Treatment:** Radioactive sources like cobalt-60 or targeted isotopes are utilized in radiation therapy to destroy cancer cells while minimizing damage to healthy tissues.
- 3. Thyroid Disorders:** Radioactive iodine-131 treats hyperactive thyroid glands and thyroid cancer by selectively damaging thyroid tissue.

AGRICULTURE

1. Soil and Crop Studies:

Radioisotopes help track nutrient uptake and soil erosion, enhancing agricultural productivity and soil management.

2. Pest Control:

Isotopes like cobalt-60 can be used to sterilize insects, reducing pest populations and protecting crops.

INDUSTRIAL FIELDS

1. Quality Control: Radioisotopes inspect welds, detect leaks, and assess material thickness in industries like construction and manufacturing.

2. Oil and Gas: Isotopes are used in reservoir studies to determine fluid flow, identify well locations, and optimize oil and gas production.

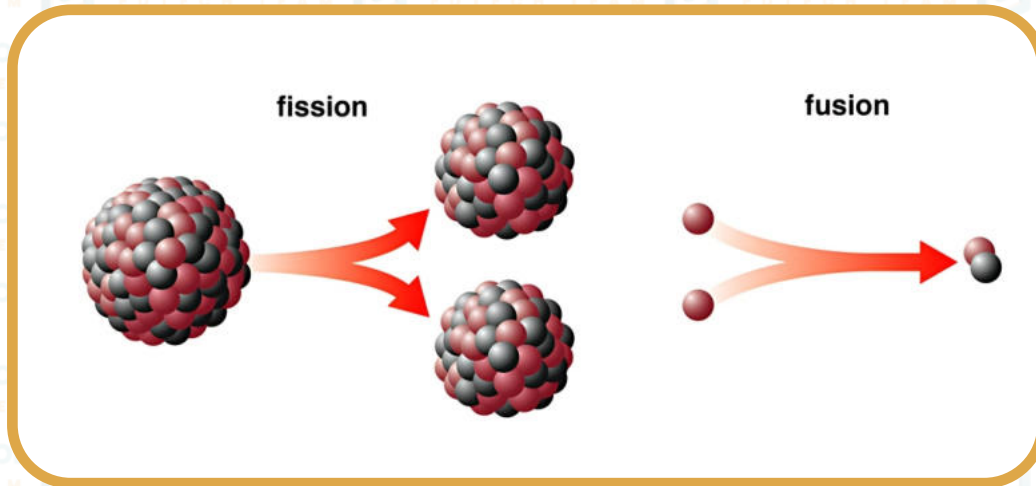
FISSION AND FUSSION

NUCLEAR FISSION

Nuclear fission is the splitting of a heavy atomic nucleus into two or more smaller nuclei, releasing a significant amount of energy.

Example:

An example is the fission of uranium-235 in nuclear reactors, producing energy and additional neutrons.



NUCLEAR FUSION

Nuclear fusion is the process of combining two light atomic nuclei to form a heavier nucleus, accompanied by a release of energy.

Example:

An example is the fusion of hydrogen isotopes (deuterium and tritium) in the sun, generating heat and light energy through a chain reaction.

RADIATION HAZARDS

Radiation hazards refer to potential risks posed by exposure to ionizing radiation, which can harm living organisms and cause damage to materials. These hazards include:

- 1. Health Effects:** Ionizing radiation can damage cells, leading to increased risk of cancer, genetic mutations, and other health problems.
- 2. Contamination:** Radioactive materials can contaminate surfaces, air, water, and food, posing both external and internal exposure risks.
- 3. Acute Radiation Syndrome:** High doses of radiation over a short period can lead to acute radiation sickness, causing nausea, skin burns, and even death.

SAFETY MEASURES

To mitigate radiation hazards, various safety measures are essential:

1. Minimize exposure time, maximize distance from the source, and use appropriate shielding materials like lead or concrete.
2. Use proper PPE, such as lead aprons, gloves, and eye protection, to reduce exposure to radiation.
3. Regularly monitor radiation levels using dosimeters and radiation detectors to assess exposure and ensure safety.
4. Design facilities and equipment with proper shielding to contain radiation and minimize exposure to personnel.
5. Provide comprehensive training to workers on radiation hazards, safety protocols, and emergency procedures.

6. Follow national and international regulations, guidelines, and exposure limits to ensure safe practices.
7. Implement medical surveillance programs for workers exposed to radiation to monitor their health and detect any potential issues early.
8. Develop emergency response plans, evacuation procedures, and communication protocols in case of radiation incidents.

